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Advanced Optimization for Big Data Streams with Quantum Insights for Real-time Big Data Analytics

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KEYWORDS

quantum-inspired computing; real-time stream processing; adaptive resource allocation; privacy-aware analytics

ABSTRACT

Big data analytics encounters scalability, latency, and privacy challenges, especially within real-time streaming contexts. We propose the Privacy-Aware Quantum Stream (PAQS), a distributed framework inspired by quantum principles, to overcome these obstacles. PAQS utilizes quantum superposition to effectively represent high-dimensional data, quantum entanglement for sophisticated correlation analysis and anomaly detection, and federated learning combined with homomorphic encryption to maintain privacy without compromising performance. The adaptive switching mechanism balances quantum-inspired and classical processing according to sensitivity and dimensionality criteria. Experiments are conducted on three datasets—OpenStreetMap, MIMIC-III, and KITTI, which show significant improvements: a throughput of 2.53 TB/sec, a 60 % reduction in latency, an anomaly detection accuracy of 92.3 %, and an 85.4 % decrease in privacy violations when compared to baselines. These findings validate that PAQS provides consistent, secure, and scalable real-time analytics, positioning it as a strong solution for smart cities, healthcare, and autonomous transportation applications.

1. Introduction

In today's technological ecosystems, real-time data analytics has become a fundamental pillar that powers vital applications in various fields, including industrial automation, smart cities, autonomous vehicles, and healthcare monitoring. These systems handle enormous amounts of ongoing data

produced by sensors, IoT devices, and human interactions (Mohbey, 2019). The main goal is to extract actionable insights in almost real-time to guide decisions and actions that directly influence operational efficiency, safety, and user pleasure. However, there are many impediments to attaining such high-performance real-time analytics in dynamic and resource-intensive contexts. As data streams grow in volume and velocity, scalability becomes a significant bottleneck, particularly during peak hours or system outages[1]. Reducing latency is essential because even a little processing delay can result in poor choices or lost opportunities, such as a delayed emergency response. In addition, many systems have trouble adapting to changing data stream patterns, such as abrupt changes in user behaviour or environmental circumstances. Finally, reaching practical decision-making under multi-objective constraints, which involves striking a balance between resource utilization, efficiency, and fairness, remains challenging.

Emerging as a revolutionary technology, quantum computing can solve the intricacies of big data analytics that traditional computing finds challenging to tackle effectively. Fundamentally, quantum computing uses the concepts of entanglement and superposition to process data at unprecedented speeds and scales. Because of these features, quantum systems are especially well-suited to handling the multidimensional datasets encountered in big data settings. With exponential speedups for certain databases and cryptographic problems, quantum algorithms such as Grover's (Khanal et al., 2023) and Shor's (Avila et al., 2017) can completely transform data processing and insight extraction. Quantum bits, also known as qubits, vary from classical bits because they can exist in multiple states concurrently, allowing for massive parallel processing. Because of this characteristic, quantum computing is well-suited for big data analytics. With progress in the field, algorithms and higher qubit fidelity improvements should boost the usability of quantum computing, turning it into an essential asset for companies dealing with the rapidly expanding data surge. Ongoing research and cooperation between quantum computing and big data analytics will be crucial in achieving this potential, paving the way for a new era of computational strength and problem-solving ability.

Despite these developments, big data applications for quantum computing still face considerable obstacles. Since complex error correction techniques and cryogenic settings are needed to construct and maintain stable quantum systems, the early stages of quantum hardware present challenges. The scalability and dependability of quantum solutions for large-scale data processing are currently limited by the inability to maintain the coherence and entanglement required for effective quantum computation over extended periods. Innovative solutions that overcome conventional systems' computational and architectural constraints and improve their capacity for real-time, context-aware, optimal decision-making are essential to meet these difficulties. This paper presents the Privacy-Aware Quantum Stream (PAQS). This quantum-inspired framework combines sophisticated optimization methods with scalable, distributed systems to address these issues and provide revolutionary advancements in real-time analytics. The major contributions of PAQS are:

- With quantum superposition-based data representation, reducing computational complexity with effective representation of high-dimensional datasets is ensured in PAQS.
- The privacy-preserving federated learning component of PAQS emphasizes the non-exposure of data. The homomorphic encryption supports secure model aggregation.

[1] https://www.netscribes.com/data-engineering-challenges-in-scaling-real-time-analytics-solutions/

- The anomaly detection is achieved in PAQS by deploying quantum entanglement principles for advanced correlation analysis.
- The experimental demonstration of PAQS on three datasets, each from diverse domains, exhibits significant generalizability across diverse domains. The stream processing suggests high transfer efficiency.

The remainder of the paper is organized as follows: Section 2 reviews recent literature on related topics, and Section 3 details the methodology. In Section 4, the experimental setting and the demonstration results are discussed. Finally, in Section 5, future work is described.

2. Related Work

In this section, we discuss recent and relevant literature citing the advancements in big data over the past decade, especially using quantum algorithms. The COVID-19 epidemic has highlighted the significance of big data. Ma et al. (2022) proposed an intelligent network information management system to address the management problem under big data and cloud computing. Mohbey and Kumar (2022) studied the impact of predictive analysis on cloud computing. Numerous studies have delved into its uses in various fields, especially in economic recovery (Shi et al., 2022), health monitoring (Knoppers and Thorogood, 2017; Asri et al., 2015), and crisis management (Qadir et al., 2016). Present methods for big data analytics encounter essential constraints that hinder their efficiency in managing the rapid increase. Conventional computing architectures face challenges with computational bottlenecks when handling high-dimensional datasets, frequently leading to considerable latency and resource usage. Although distributed computing and parallel processing provide some solutions, they still encounter fundamental scalability limitations and typically necessitate significant infrastructure expenditures (Taherdoost, 2024). Cloud-based solutions are flexible but face bandwidth restrictions and privacy issues during data transfer and processing (Sandhu, 2021). Machine learning methods, especially deep learning frameworks, require significant computational power and training duration, rendering real-time analytics difficult.

Quantum computing surfaces are a promising answer to these constraints due to their distinct parallel processing and optimization abilities. Via quantum superposition, a quantum system can process several data states at once, which could provide exponential acceleration for certain large data tasks. Quantum entanglement allows for improved correlation analysis within extensive datasets, whereas quantum tunnelling can optimize intricate analytical models more efficiently than traditional algorithms. Researchers are exploring the potential of quantum computing to bridge the gap between theoretical research and practical implementations. Ashraf et al. (2025) proposed a methodology in which a neural network framework used differential privacy to identify data leakage in the context of wireless network resource allocation. This framework utilized both linear and nonlinear neural network mappings to simulate, train, and assess its performance compared to conventional methods. Al-Shiakhli (2019) outlined big data analytics methods and resources in many sectors, emphasizing the significance of these tools in decision-making and pointing out gaps in the literature. Shaikh and Ali (2016) reviewed the role of quantum computing in big data analytics and highlighted its advantages over traditional classical methods and the impending challenges. The studies of Yang et al. (2017) and Gautam and Chatterjee (2020) highlighted the potential of quantum algorithms in big data processing. Their studies focused on the potential integration of quantum computing into cloud-based big-data

analytics, the anticipated accelerated processing, data encryption, and the impending challenges. Liu and Jiang (2024) proposed a quantum data centre (QDC) that deployed quantum random access memory (QRAM). QDC demonstrated significant improvements in efficiency, security, and precision.

Ramezani et al. (2020) assessed the speed and complexity benefits of employing quantum machines and provided an overview of the current knowledge in applying ML on QC. Khurana (2022) critically examined the three applications of QC: Quantum Approximate Optimization Algorithm (QAOA), Quantum Key Distribution (QKD), and Quantum Machine Learning (QML). The study also addressed issues such as qubit decoherence, quantum error correction, and integration of quantum and classical systems. QC has also been studied in the context of machine learning algorithms. Bishwas et al. (2018) proposed a method for multiclass classification in massive data using all-pairs quantum support vector machines (SVMs). In addition to offering a framework for future research into quantum machine learning applications, the study also demonstrated how quantum SVMs can improve classification accuracy and speed. According to a comparative study of classical-to-quantum mapping methods by Rath and Date (2024), efficient data embedding is crucial for maximizing performance in quantum machine learning tasks.

Agarwal and Alam (2021) explored the convergence of quantum computing and machine learning, illustrating how quantum algorithms could enhance data processing. Their study indicated that machine learning models enhanced by quantum technology exceed conventional methods in accuracy and processing speed. In their extensive review, Osaba et al. (2022) highlighted how quantum computing could address routing problems and highlighted its effectiveness in optimization tasks involving big data. The authors suggested avenues for future research in quantum routing algorithms. Havlíček et al. (2019) investigated supervised learning strategies using quantum feature spaces, achieving performance metrics that were superior to traditional approaches across various applications, such as big data categorization tasks. Quantum versions of conventional algorithms, such as k-means clustering, have been proposed to handle large data clusters (Tsai et al., 2015). The development of quantum-enhanced data analysis tools, which seek to augment the statistical approaches employed in big data analytics, is another significant achievement (Wang, 2022). Andreou et al. (2025) introduced a hybrid quantum-classical model that combines quantum-inspired optimization with cloud-edge AI systems to improve automation, efficiency, and sustainability. It integrates methods like QUBO, QAOA, and federated learning, minimizing computational complexity, energy usage, and data transfer costs.

3. Methodology

PAQS is a distributed quantum-inspired system for big data analysis that tackles scalability, efficiency, and privacy issues. It starts with data representation through quantum superposition for parallel processing and utilizes preprocessing techniques such as normalization and noise reduction. Distributed quantum-inspired algorithms utilize entanglement for correlation assessment, clustering, and anomaly identification. Federated learning (FL) maintains privacy by training models on local edge devices, using secure aggregation for global optimization while keeping raw data confidential. The concluding stage encompasses real-time analysis through Kafka pipelines, quantum-inspired decision-making systems, and visualization instruments. The method combines advanced strategies to enhance practical applications in intelligent urban environments, healthcare, and self-driving transportation. Figure 1 depicts the PAQS framework, and Algorithm 1 represents the entire PAQS process stepwise. Table 1 summarizes the different symbols used.

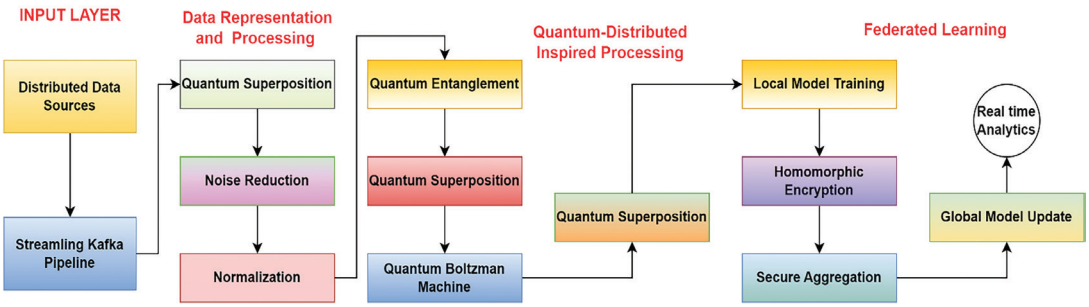


Figure 1. PAQS framework: three-phase architecture for privacy-preserved quantum-inspired data processing

Algorithm 1: Privacy-Aware Quantum Stream (PAQS) algorithm

Input: Distributed dataset D, privacy constraint π, computational resource R, Quantum Parameters QP

Output: Processed Insights I, privacy-preserved model M

1. For each data point $x \in D$:
   $\alpha, \beta \leftarrow$ QuantumSuperposition(x)
   Normalize(α, β)
   Apply Denoising(D)
   Perform Min-Max Normalization(D)
2. For each node $n \in D$:
   $E(A,B) \leftarrow$ QuantumEntanglement(Local_Data) // Ref Equation 1
   Correlations[n] $\leftarrow E(A,B)$
3. Initialize Quantum Centroids
   While not converged:
   $d_Q \leftarrow$ QuantumDistanceMeasure() // Ref Equation 2
   Assign data points to the nearest centroid.
   Update Centroids
4. Initialize Quantum Boltzmann Machine
   Calculate Energy Function $E(x)$ // Ref Equation 3-4
   $P(anomaly) \leftarrow$ ProbabilisticThresholding($E(x)$)
5. Adaptive Switching between classical and quantum-inspired switching // Ref Equation 5-7
6. **For** each edge device
   Train Local Model M_{local}
   Encrypt Model Updates
7. Homomorphic Encryption of Local Updates
8. Aggregate Encrypted Updates
9. Compute Global Model Parameter
10. Process Streaming Data, generate insights I

Table 1. Symbol table summarizing the various notations used

Symbol	Meaning
x_i	Data point
ψ	Quantum state
α, β	Probability amplitudes
ρ_{AB}	joint density matrix
$d_Q(x_i, x_j)$	quantum-inspired distance between x_i and x_j
k	Clusters
q	quantum distance parameter
τ	Controlling factor
D_{data}	data dimensionality
θ_r	Adaptation threshold
l	Loss function
η	Learning rate
b_i	Quantum Boltzmann machine bias

3.1. Data Representation and Preprocessing

The raw datasets were taken from distributed sources and were efficiently preprocessed into high-quality, consistent data for downstream analytics. Utilizing quantum superposition decreases computational complexity by allowing parallel data representation. We encode each data point x_i as a quantum state $|\psi\rangle$ as: $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$where α and β are probability amplitudes owing to $|\alpha|^2 |\beta|^2 = 1$. Next, a noise reduction using wavelet transformers is performed to remove the inconsistencies $f_{cleaned(x)} = \Sigma_{i=1}^{n} f(x_i) \cdot W_t(x_i)$ where $W_t(x_i)$ is a wavelet coefficient. Then, normalization ensures scaling, such as: $x_{norm} = \dfrac{x - \min(x)}{\max(x) - \min(x)}$. The process returns clean, normalized, quantum-represented data.

3.2. Distributed Processing with Quantum-Inspired Algorithms

The preprocessed data from the previous step is distributed across multiple nodes. In this step, we perform quantum entanglement for correlation analysis as per Equation (1) to enhance pattern discovery and to measure inter-node data connections in distributed quantum-inspired systems

$$E(A, B) = T_r [\rho_{AB} * (\sigma_A \otimes \sigma_B)] / (\|\sigma_A\| * \|\sigma_B\|) \quad (1)$$

Where, ρ_{AB} is the joint density matrix, σ_A and σ_B observables for systems A and B, $T_r[]$ is the matrix trace operator, $\otimes$ is the tensor product, and $\|.\|$ is the norm of matrices. Next, using the quantum-inspired distance $d_Q(x_i, x_j)$ e (Equation (2)) between the data points (x_i, x_j), we perform the clustering iteratively to construct k clusters.

$$d_Q(x_i, x_j) = \sqrt{\sum \left(\alpha_k * \left| x_{i_k} - x_{j_k} \right|^q \right)} \quad (2)$$

Where, α_k is the quantum weight factor and q quantum distance parameter. Besides this, we adopt quantum Boltzmann machines to flag anomalies using the energy function $E(x)$ as represented in Equation (3)-(4).

$$E(x) = -\Sigma(W_{ij} * s_i * s_j) - \Sigma(b_i * s_i) \tag{3}$$

$$P(anomaly) = 1 / (1 + exp(E(x)/\tau)) \tag{4}$$

Where, s_i and s_j are the state variables, τ is the controlling factor, and b_i and W_{ij} are the learned parameters. Next, we apply the dynamic computational resource allocation that alternates between quantum-inspired and classical processing based on data dimensionality D_{data}, and sensitivity scores S_ρ. When D_{data} and S_ρ exceeds the threshold, processing shifts to quantum-inspired algorithms. S_ρ uses pattern matching for risk evaluation. The mathematical process is explained in Equations (5)-(7).

$$D_{data} = \frac{\eta_f \cdot \eta_z}{\eta_f + \eta_z} \tag{5}$$

$$\eta_z = \sum_{i=1}^{\eta_f} I \cdot \lambda_i > 0.1 \tag{6}$$

$$S_\rho = \min_i \left(\frac{\sum_{i=1}^{p} \omega_i \cdot m_i}{B}, 1.0 \right) \tag{7}$$

Where, η_f represents the number of features, η_z denotes the effective dimensions computed from eigenvalues λ_i of the correlation matrix, ω_i is the sensitivity weights, m_i denotes the pattern matches, p is the total number of patterns, and B is the batch size. Based on m_i, ω_i and B, we compute the S_ρ. If the S_ρ is greater than the τ then we invoke a quantum-inspired algorithm; else, the system relies on classical processing. It ensures that the quantum algorithm is used only in scenarios where there is a substantial benefit. Further, we assess the system load, resource availability, and dataset domain to set a threshold θ_r. The composite score $C_s = 0.4\, D_{data} + 0.3 S_\rho$. When $C_s > \theta_r$ the algorithm adopts quantum processing; otherwise, classical processing is used.

3.3. Federated Learning for Privacy Preservation

In this phase, we implement federated learning on the clustered data using local model updates from dispersed nodes. It guarantees decentralized computation, protects private information, and complies with privacy laws. Each edge device is modelled locally, $L(\theta) = \frac{1}{m} \sum_{i=1}^{m} l \cdot (f(x_i; \theta), y_i)$ where l is the loss function, and (x_i, y_i) are input-output pairs. Using homomorphic encryption, we securely aggregate the updates $c_k = \prod_{i=1}^{N} g^{m_{ik}} \bmod p$ where m_{ik} are model updates. The aggregated updates refine the global model with η learning rate as $\omega_t = \omega_{t-1} - \eta \nabla F(\omega_t)$. With this, we obtain the global model without the

exchange of raw data. Each node allocated 10 % of its local dataset for validation before submitting encrypted updates, ensuring that only models with strong generalization capabilities took part in the global aggregation process. This supplementary validation phase added minimal overhead, as it required lightweight forward evaluations and did not significantly influence execution time.

Streaming platforms and quantum-inspired decision-making algorithms, which process data dynamically as it is generated, are integrated into this approach to provide real-time analytics. Apache Kafka was first used to handle real-time data ingestion. It gathers, analyses, and distributes real-time data from multiple sources. Next, the distribution of data processing between centralized cloud systems and edge devices (near the data source) lowers latency and guarantees real-time responsiveness. The platform provides real-time analytics appropriate for dynamic, large-scale environments by fusing streaming infrastructure, distributed quantum-inspired computation, and sophisticated decision-making algorithms.

The algorithm starts by converting each data point into a quantum superposition state, allowing for concurrent multidimensional representation via probability amplitudes α and β. It is subsequently subjected to thorough denoising and normalization procedures. In step 2, Quantum entanglement forms intricate associations among distributed nodes, generating a complex inter-node relationship matrix. Then, it utilizes quantum-inspired clustering methods, applying unique distance metrics and probabilistic centroid initialization to cluster data points while ensuring computational adaptability effectively. A quantum Boltzmann machine (step 4) is employed for anomaly detection, computing energy functions that probabilistically recognize and highlight atypical data patterns with great accuracy. Next, the federated learning aspect allows for localized model training on edge devices, where encrypted model updates are collected using homomorphic encryption, guaranteeing total privacy protection throughout the global model optimization procedure.

4. Experimentation and Results

This section discusses the benchmarks, evaluation measures used, and the experimental setup required for the demonstration. Next, we elaborate on the results obtained and discuss the hyperparameters involved.

4.1. Experimental Setting

The software ecosystem encompasses Python 3.9+, which utilizes PennyLane for quantum computing simulations, TensorFlow Quantum to tackle multi-objective decision issues, Apache Spark for distributed data processing, and Dask for parallel computing tasks. Extra prerequisites encompass Kafka for stream processing, Kubernetes for managing containers, and the specialized quantum simulation framework Cirq to facilitate quantum-inspired algorithm implementations.

4.2. Hyperparameter Settings

The batch size is set to 512 to maintain model convergence, and the learning rate η is 0.001 set after performing a grid search on [0.0001, 0.001, 0.0025, 0.01]. We adopt a 16-bit encryption base that affects the security level of homomorphic encryption and balances the computational cost. Another critical factor is the number of quantum states (Q) set to 32. A large value of Q diminishes resource utilization. In real-time data processing, a window size of 10 seconds is effective without burdening the system. The quantum probability amplitudes $\alpha = 0.63$ and $\beta = 0.7$. Quantum Boltzmann machine bias, $b_i = 0.45$ and W_{ij}, which controls anomaly detection and affects pattern recognition capabilities, is set at

0.562. The quantum clusters k were set to 6 for MIMIC-III, 8 for KITTI, and 10 for OpenStreetMap. Threshold for Switching, θ_r, is empirically tuned and θ_r=0.65±0.15 provided the trade-off between performance and resource utilization. Further, local epochs were set to 5 in federated learning. We have used the Adam optimizer.

4.3. Datasets Description

The PAQS has been validated on three datasets, which are detailed below.

- OpenStreetMap Data[2]: This dataset contains geographical data demonstrating scalability and real-time analytics in smart city scenarios.
- MIMIC-III Clinical Database[3] includes de-identified medical records to assess customized healthcare apps using privacy-preserving methods.
- KITTI Dataset[4]: This dataset offers sensor data for activities such as object identification and tracking, which are used to evaluate how well autonomous transportation systems operate.

4.4. Evaluation Metrics

- Throughput: It is quantified by the volume of data handled each second, using Apache Kafka for streaming data and Spark for batch processing. The system tracks the number of records or data points handled per second. This metric indicates the system's capability to manage data spikes, guaranteeing scalability and resilience under high loads.
- Latency: The time this framework requires to analyze and deliver insights is essential. Various elements, such as the amount of incoming data, the effectiveness of the quantum-inspired algorithms, network latency, and the hardware's processing power, may influence latency.
- Anomaly Detection: This metric assesses the system's capacity to accurately recognize abnormal patterns by contrasting its outputs with the ground truth labels found in the datasets. Ground truth labels are obtained via manual tagging or recognized anomaly standards, guaranteeing dependability.
- Data Security: It is evaluated according to the percentage decrease in privacy infractions.
- Computational Resource Utilization Ratio (CRUR): It measures the optimization of computational resource usage. It evaluates resource utilization efficiency.
- Convergence Time Reduction Percentage (CTRP) measures the model's convergence speed.
- Privacy Breach Resistance Index (PBRI): It assesses the robustness of the model against data breaches and subsequently evaluates the model's capability in protecting sensitive information.
- Scalability Performance Index (SPI): It measures how far the system can maintain its performance under the load conditions.
- Domain Transfer Efficiency Ratio (DTER): This is useful for measuring cross-domain generalizability across diverse computational environments.

[2] https://www.openstreetmap.org/
[3] https://mimic.physionet.org/
[4] http://www.cvlibs.net/datasets/kitti/

4.5. Results

Every metric exemplifies transformative gains in handling intricate, large-scale processing problems while maintaining strict efficiency and privacy standards.

4.5.1. Benchmarks

Experimental analysis of the MIMIC-III clinical database is presented in Table 2, which suggests that PAQS consistently exceeds conventional big data analytics platforms, showcasing higher performance in throughput, reaching 2.53 TB/sec versus Apache Spark's 1.82 TB/sec and significantly outpacing Hadoop's 0.94 TB/sec. Latency metrics highlight the framework's effectiveness, featuring the fastest response time of 120.45 ms, which is substantially better than Hadoop's 450.89 ms and centralized federated learning's 280.67 ms. The most notable enhancement is seen in data security, as the framework reaches an 85.42 % decrease in privacy violations, significantly surpassing centralized federated learning's 50.13 % decrease. The anomaly detection accuracy further supports the method, achieving 92.34 % in contrast to Apache Spark's 78.29 % and Hadoop's 65.12 %, underscoring quantum-inspired algorithms' advanced pattern recognition abilities. Table 3 depicts the comparison of benchmark metrics with the different datasets.

Table 2. Performance comparison of PAQS with baselines

Metric	**PAQS**	**Apache Spark**	**Hadoop**	**Centralized Federated Learning**
Throughput (TB/sec)	2.53	1.82	0.94	1.21
Latency (ms)	120.45	300.76	450.89	280.67
Data Security (%)	85.42 %	N/A	N/A	50.13 %
Anomaly Detection (%)	92.34	78.29	65.12	85.47

Table 3. Comparison of benchmark evaluation metrics for different datasets

Metric	**Throughput (TB/sec)**	**Latency (ms)**	**Data Security (%)**	**Anomaly Detection (%)**
OpenStreetMap	2.1	145.2	82.1	89.2
MIMIC-III	2.53	120.45	85.47	92.34
KITTI	2.4	135.8	83.7	90.1

4.5.2. Efficiency Metrics

Table 4 demonstrates the efficiency analysis conducted on the MIMIC-III clinical database. The results indicate that PAQS has improved by 40.3 % in CRUR over Hadoop. Primarily because its sophisticated algorithm lowers computational costs, decreases energy usage, and enhances hardware effectiveness. CTRP also exhibits a similar trend with a 54.8 % improvement as the quantum-inspired framework ensures the exploration of different solution spaces that accelerate the training process, thereby reducing computational time. Further, this federated learning method and sophisticated encryption methods establish multiple layers of security measures. Through the decentralization of data processing and the strong privacy protection measures, the framework significantly reduces the potential

risks of data breaches. Hence, PBRI provides a 31.9 % improvement over centralized systems. The improved load balance and quantum representation showed a 61.2 % improvement in SPI over Hadoop. DTER statistics suggest unprecedented adaptability across diverse computational domains.

Table 4. Efficiency metrics comparison with baselines

Metric	Frame	Apache Spark	Hadoop	Centralized Federated Learning
CRUR	0.87	0.62	0.48	0.55
CTRP	0.65	0.42	0.31	0.38
PBRI	0.92	0.68	0.53	0.71
SPI	0.79	0.56	0.41	0.49
DTER	0.73	0.51	0.37	0.44

4.5.3. Validation

Each experiment was evaluated 10 times under identical experimental settings across all three datasets. Table 5 demonstrates the validation results regarding mean values, standard deviation, and 95 % confidence interval (CI), selected for the key performance metrics, i.e., throughput, latency, accuracy, and data security. The confidence interval (CI) for throughput is notably narrow (2.48–2.58 TB/sec), which signifies that PAQS reliably upholds a high data-handling capacity with minimal variation across different runs.

Table 5. Statistical validation of PAQS performance metrics over 10 experimental runs

Metric	Dataset	Mean±Std. Dev	95 % Confidence Interval (CI)
Throughput (TB/sec)	MIMIC-II	2.53±0.07	2.48-2.58
Latency (ms)	MIMIC-II	120.45 ±3.20	118.0-124.0
Anomaly detection	KITTI	90.1±1.1	89.0-91.5
Data Security	OpenStreetMap	82.1±1.4	80.5-83.7

In contrast, the CI for latency is broader (118–124 ms). This variability is anticipated, as transient system-level dynamics, including queueing delays, spikes in streaming workloads, and network jitter, inherently influence latency. Put differently, while throughput indicates the overall system capacity that remains stable, latency reflects real-time responsiveness, where even minor environmental changes can have a significant effect. A similar pattern is evident in anomaly detection and data security: both demonstrate narrow intervals, which confirms that PAQS's quantum-inspired algorithms deliver consistent accuracy and privacy protection across various datasets. Notably, none of the intervals intersect with baseline system results, strengthening the assertion that the observed enhancements are statistically significant and not merely the result of random chance.

From a deployment standpoint, these results imply that PAQS can be depended on for reliable performance in high-throughput settings, while its latency profile—despite being naturally variable—stays comfortably within acceptable limits for applications requiring real-time decision-making.

4.5.4. Impact of Semi-Structured or Noisy Data

For semi-structured or noisy data, which frequently lead to false positives in anomaly detection, PAQS tackles this challenge by utilizing wavelet-based denoising and normalization preprocessing before conducting correlation analysis, thereby significantly diminishing noise sensitivity. These integrated strategies illustrate that PAQS is effective and resilient to practical deployment challenges, positioning it as a strong candidate for large-scale, real-time big data environments.

4.5.5. Discussion

The experimental findings emphasize various key factors that notably impacted the performance of PAQS. The quantum state parameter Q=32 was optimal after evaluating values from 8 to 64 since higher values reduced processing efficiency and heightened resource usage. The adaptation threshold θ_r kept at 0.65±0.15 was essential for balancing resource allocation between quantum and classical, while specific adjustments $\left(\theta_r^{health} = 0.55, \theta_r^{city} = 0.60, \theta_r^{iot} = 0.75\right)$ enhanced performance in various applications. The learning rate η = 0.001 was identified as optimal via grid search, striking a balance between convergence speed and stability. The slight differences in latency (Table 5) are anticipated results of temporary system-level dynamics, including fluctuations in streaming workloads and network jitter. These differences do not compromise the reliability of other metrics, which have remained consistent throughout repeated executions. Significant challenges involved sustaining quantum coherence while performing entanglement-based correlation analysis. Particularly as data dimensionality rises, the trade-off between safeguarding privacy and ensuring computational efficiency is balanced. The computation of the privacy score sometimes resulted in false positives with semi-structured data, necessitating manual adjustments to the threshold. The system demonstrated notable sensitivity to changes in batch size, where 512 offered the optimal trade-off between processing efficiency and memory use. These findings indicate that although PAQS provides considerable advancements compared to conventional methods, meticulous parameter adjustment is essential for achieving the best performance in various fields.

While computational overheads are prevalent in privacy-preserving frameworks, PAQS effectively addresses this concern by implementing parallelization and batching strategies. This ensures that the extra costs associated with homomorphic encryption remain manageable, even when faced with substantial workloads. In a similar vein, although hyperparameters such as batch size, adaptation threshold (θ_r.), and quantum state parameters are recognized for their impact on performance, PAQS employs a systematic tuning approach, including grid search and sensitivity analysis, to stabilize results across various datasets and operating conditions.

5. Conclusion

This paper presents a distributed quantum-inspired analytics framework, PAQS for big data, successfully tackling scalability and privacy issues in contemporary big data systems. Utilizing quantum-inspired superposition for optimized data representation and entanglement for comprehensive correlation analysis, combined with federated learning to maintain privacy, PAQS shows considerable improvements in computational efficiency and data protection. The results indicate an enhancement in anomaly detection precision and a decrease in privacy violations compared to conventional systems. This allows for more efficient use of computational resources, reducing energy consumption and infrastructure costs.

It ensures that raw data remains decentralized and protected, significantly mitigating risks associated with traditional centralized data processing architectures. The system can seamlessly transition between smart city infrastructure, healthcare analytics, and autonomous transportation systems without significant architectural modifications by abstracting core quantum-inspired principles. The results underscore the potential of quantum-inspired computing as a paradigm-shifting approach to big data analytics, offering a compelling alternative to conventional computational methodologies. This carries significant consequences for practical applications, such as improving urban resource management in smart cities, offering personalized and secure healthcare advice, and advancing real-time decision-making in autonomous transport. Future uses could expand into disaster response, live financial analysis, and secure military functions, where scalable and privacy-centered big data analytics are essential.

6. Ethics

6.1. Conflict of Interest

The authors have no competing interests to declare that are relevant to the content of this article.

6.2. Contributors

M. Acharya: Conceptualization, Methodology, Software, Data curation. **K.K. Mohbey:** Supervision, Writing – review & editing.

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